

A Short Guide to Lemma C.1

Weighted root-fidelity bounds

Abstract

Lemma C.1 converts root fidelity into a product of weighted moments. Its first part is an operator inequality. Its second part combines that inequality with classical Cauchy–Schwarz for a classical–quantum state. The weights used later are chosen so that one moment is compact and can pass to a weak limit, while the remaining factors are exactly the classical and orbital Gaussian affinities. This note explains the regularized inverse, the two Cauchy–Schwarz steps, and the role of the lemma in Proposition C.5.

1 Why a weighted bound is useful

For positive trace-class operators R and T , their root fidelity is

$$B(R, T) = \|R^{1/2}T^{1/2}\|_1.$$

Appendix C must control this nonlinear quantity along sequences that converge only after testing against compact observables. Fidelity itself is not one of those linear tests. Lemma C.1 replaces it by a moment

$$R \longmapsto \text{Tr}(RW)$$

for a bounded positive compact operator W . This moment does pass to the limit supplied by Lemma C.2.

The inverse weight is paid on the fixed target:

$$B(R, T)^2 \leq \text{Tr}(RW) \text{Tr}(TW^{-1}).$$

Thus W should make the first factor continuous in the available topology and the second factor explicitly computable. The witness W_γ used in Appendix C has exactly these two properties.

2 The regularized inverse

The weight W is bounded, positive, and injective, but it need not be bounded below. Its inverse may therefore be unbounded. The expression $\text{Tr}(TW^{-1})$ is defined through bounded regularizations:

$$\text{Tr}(TW^{-1}) := \lim_{\varepsilon \downarrow 0} \text{Tr}(T(W + \varepsilon I)^{-1}). \quad (1)$$

As ε decreases, the positive operators $(W + \varepsilon I)^{-1}$ increase in the spectral order. Hence the limit exists in $[0, \infty]$. The operator inequality is useful when this limit is finite.

Injectivity rules out a genuine zero eigenspace. It does not make W^{-1} bounded, and that is why the definition in (1) is needed. For the later thermal witness, the eigenvalues of W_γ tend to zero, so this distinction is essential.

3 Statement of Lemma C.1

Lemma C.1 (rephrased). *Let R, T be positive trace-class operators, and let W be bounded, positive, and injective. If (1) is finite, then*

$$B(R, T)^2 \leq \text{Tr}(RW) \text{Tr}(TW^{-1}). \quad (2)$$

Let $\mathbf{E}$ be a Euclidean space, let $R \in L^1(\mathbf{E}; \mathcal{T}_1(\mathcal{F}_{\text{orb}}))$ be positive, let g be a probability density, and let $\chi > 0$ be measurable. Whenever the right side is finite,

$$B(R, gT)^2 \leq \left(\int_{\mathbf{E}} \chi(y) \text{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right) \text{Tr}(TW^{-1}). \quad (3)$$

The notation gT means the classical–quantum density $y \mapsto g(y)T$. When $\dim \mathbf{E} = 0$, the classical integrals reduce to evaluation at the unique point, and (3) becomes (2).

4 Proof walkthrough

Step 1: insert a bounded inverse

Set

$$W_\varepsilon = W + \varepsilon I.$$

This operator is strictly positive and has a bounded inverse. Insert $W_\varepsilon^{1/2} W_\varepsilon^{-1/2} = I$ between the two square roots:

$$R^{1/2} T^{1/2} = R^{1/2} W_\varepsilon^{1/2} W_\varepsilon^{-1/2} T^{1/2}.$$

Schatten Cauchy–Schwarz, namely $\|AB\|_1 \leq \|A\|_2 \|B\|_2$, gives

$$\begin{aligned} B(R, T) &\leq \|R^{1/2} W_\varepsilon^{1/2}\|_2 \|W_\varepsilon^{-1/2} T^{1/2}\|_2, \\ B(R, T)^2 &\leq \text{Tr}(RW_\varepsilon) \text{Tr}(TW_\varepsilon^{-1}). \end{aligned} \quad (4)$$

The identities in the second line follow from the definition of the Hilbert–Schmidt norm and cyclicity of the trace.

Step 2: remove the regularization

The first factor in (4) satisfies

$$\text{Tr}(RW_\varepsilon) = \text{Tr}(RW) + \varepsilon \text{Tr} R \longrightarrow \text{Tr}(RW).$$

The second factor converges by the definition of $\text{Tr}(TW^{-1})$. Letting $\varepsilon \downarrow 0$ proves (2).

Step 3: apply the operator bound pointwise

For classical–quantum positive densities, root fidelity decomposes over the classical variable. Since the target density is $g(y)T$,

$$B(R, gT) = \int_{\mathbf{E}} \sqrt{g(y)} B(R(y), T) \, dy.$$

Apply (2) to $R(y)$ and T :

$$B(R(y), T) \leq \sqrt{\text{Tr}[R(y)W]} \sqrt{\text{Tr}(TW^{-1})}.$$

The target factor is independent of y , so

$$B(R, gT) \leq \sqrt{\text{Tr}(TW^{-1})} \int_{\mathbf{E}} \sqrt{g(y) \text{Tr}[R(y)W]} \, dy. \quad (5)$$

Step 4: split the classical weight

Insert $\chi(y)^{1/2}\chi(y)^{-1/2}$ into the integrand in (5):

$$\sqrt{g(y) \operatorname{Tr}[R(y)W]} = \sqrt{\chi(y) \operatorname{Tr}[R(y)W]} \sqrt{\frac{g(y)}{\chi(y)}}.$$

Ordinary Cauchy–Schwarz on $\mathbf{E}$ gives

$$\begin{aligned} & \left(\int_{\mathbf{E}} \sqrt{g(y) \operatorname{Tr}[R(y)W]} \, dy \right)^2 \\ & \leq \left(\int_{\mathbf{E}} \chi(y) \operatorname{Tr}[R(y)W] \, dy \right) \left(\int_{\mathbf{E}} \frac{g(y)}{\chi(y)} \, dy \right). \end{aligned}$$

Combining this with (5) proves (3).

5 The witnesses used in Proposition C.5

For the oscillator component, Appendix C takes

$$W_\gamma = \sum_{\mathbf{k}} \sqrt{\frac{t_{\mathbf{k}}}{t_{\mathbf{k}}^{(\gamma)}}} |\mathbf{k}\rangle\langle\mathbf{k}|.$$

It is compact and injective, and it satisfies

$$\operatorname{Tr}(\tau_p W_\gamma^{-1}) = \operatorname{Tr}(\tau_p^{(\gamma)} W_\gamma) = B_{\text{orb}}(\gamma, p).$$

For the full classical–quantum case, the classical weight is

$$\chi(y) = \sqrt{\frac{f_1(y)}{f_\gamma(y)}}, \quad \int_{\mathbf{E}} \frac{f_1(y)}{\chi(y)} \, dy = B_{\text{cl}}(\gamma, d).$$

Thus Lemma C.1 turns fidelity into one channel-dependent compact moment and two fixed affinity factors. Proposition C.3 bounds the quantum moment, while Lemma C.4 controls the trace available to it in the full case. For the orbital case, $\mathbf{E} = \{0\}$ and $\chi = 1$, so the classical factor is absent.

6 Points worth keeping straight

1. The inverse of W need not be bounded. Only its regularized expectation in the fixed target must be finite.
2. Compactness of W is not required for Lemma C.1 itself. It is required later to pass the moment through the compact-observable limit.
3. Two different Cauchy–Schwarz inequalities are used: Schatten Cauchy–Schwarz for operators and ordinary Cauchy–Schwarz for the classical variable.
4. The same statement covers the orbital case by taking the classical Euclidean space to have dimension zero.